DO NOW — RECALL

Exam-style data representation problems

Name: _____ Class: _____ Date: _____

1. What does CountAtLeast([5,7,3],5) return?

2. What does O(log n) often describe in binary search?

3. What is the sign bit for a negative 8-bit two's-complement integer?

4. Calculate $0.1000_2 \times 2^3$.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

2.

2. Model answer

Repeatedly halving the candidate interval.

3. Model answer

1.

4. Model answer

4.

Vocabulary — know these terms

bit width — Number of bits in a representation.

two's complement — Fixed-width signed binary format.

mantissa — Signed fractional significant component.

exponent — Power-of-two scaling value.

precision — Number of significant digits retained.

NEW LEARNING

Read, annotate and explain

Respect the bit width

Write positive 5 using exactly eight bits.

Invert then add one

The result is 11111011_2, representing −5 in 8-bit two's complement.

Interpret floating-point differently

0.1100_2 is fractional 0.75; multiplying by 2^3 gives 6.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

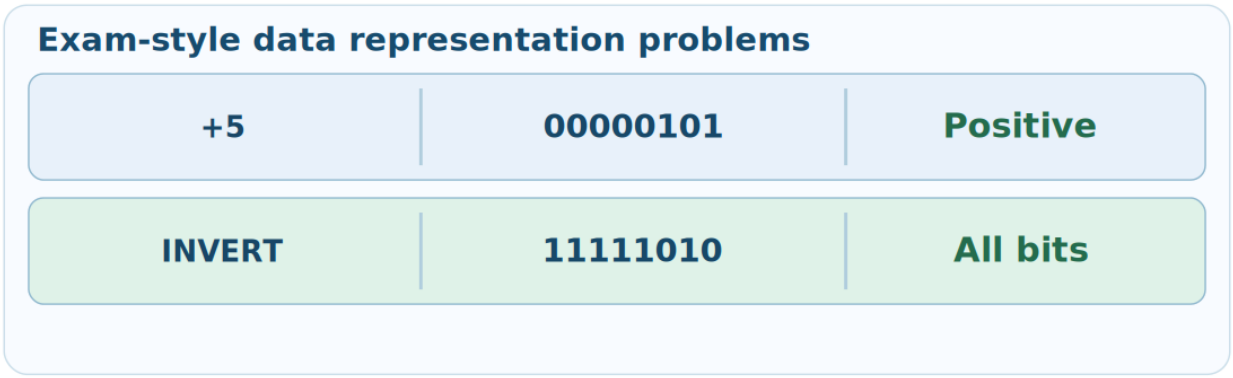
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Exam-style data representation problems: +5, 00000101, Positive

New Learning

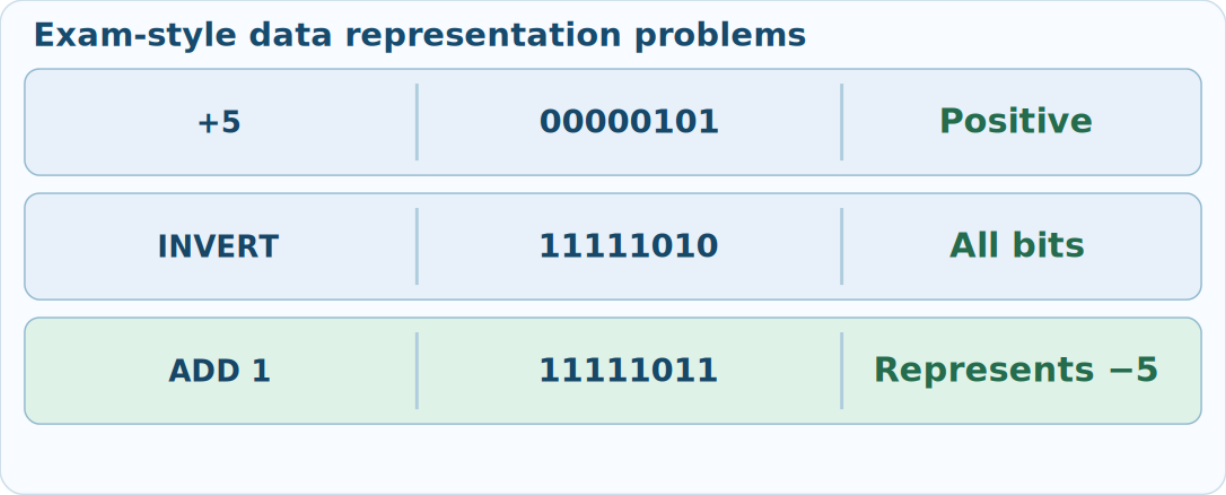


Exam-style data representation problems: +5, 00000101, Positive; INVERT, 11111010, All bits

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Exam-style data representation problems: +5, 00000101, Positive; INVERT, 11111010, All bits; ADD 1, 11111011, Represents −5

I DO — WORKED EXAMPLE

Follow the worked method

Convert −10 to 8-bit two's complement.

Predict

Write 00001010 for +10 and invert to 11110101.

Explain

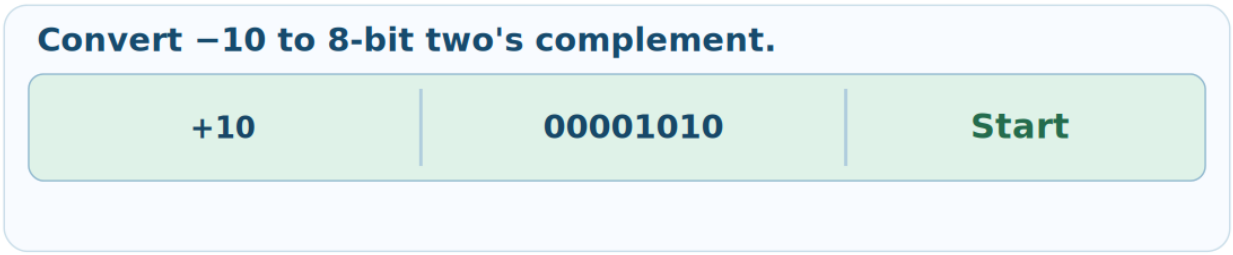
Add one: 11110110 represents −10.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Convert −10 to 8-bit two's complement.

Write 00001010 for +10 and invert to 11110101.



Convert −10 to 8-bit two's complement.: +10, 00001010, Start

Convert −10 to 8-bit two's complement.

Write 00001010 for +10 and invert to 11110101.



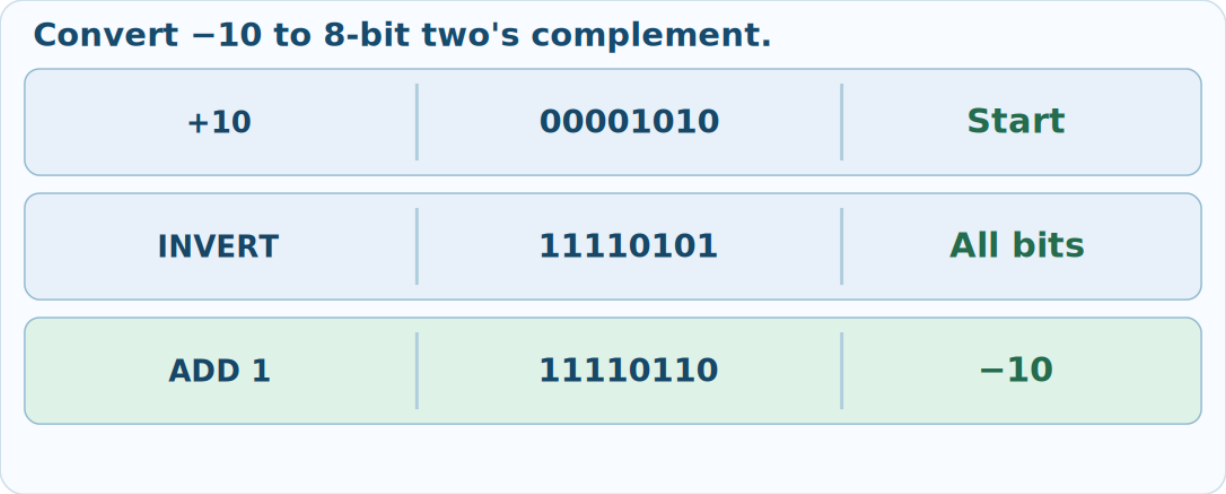
Convert −10 to 8-bit two's complement.: +10, 00001010, Start; INVERT, 11110101, All bits

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Convert −10 to 8-bit two's complement.

Write 00001010 for +10 and invert to 11110101.



Convert −10 to 8-bit two's complement.: +10, 00001010, Start; INVERT, 11110101, All bits; ADD 1, 11110110, −10

WE DO — GUIDED PRACTICE

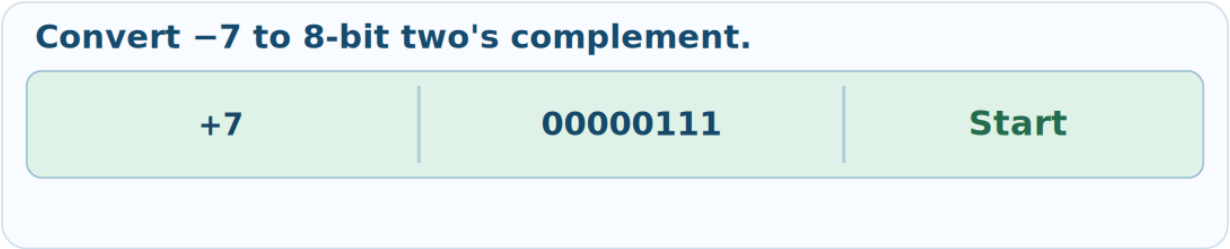
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Convert −7 to 8-bit two's complement.

Start with +7 = 00000111 and invert all bits.



Convert −7 to 8-bit two's complement.: +7, 00000111, Start

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict Start with +7 = 00000111 and invert all bits.

DISCUSS — Compare intermediate values and your reasoning. Justify one step using the relevant rule, representation or truth table.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Convert unsigned binary 00001011 to decimal.

Question 2

Encode −12 in 8-bit two's complement.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Normalise positive $0.0011_2 \times 2^5$ using a two's-complement fractional mantissa.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Eight-bit two's complement can represent decimal +128.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

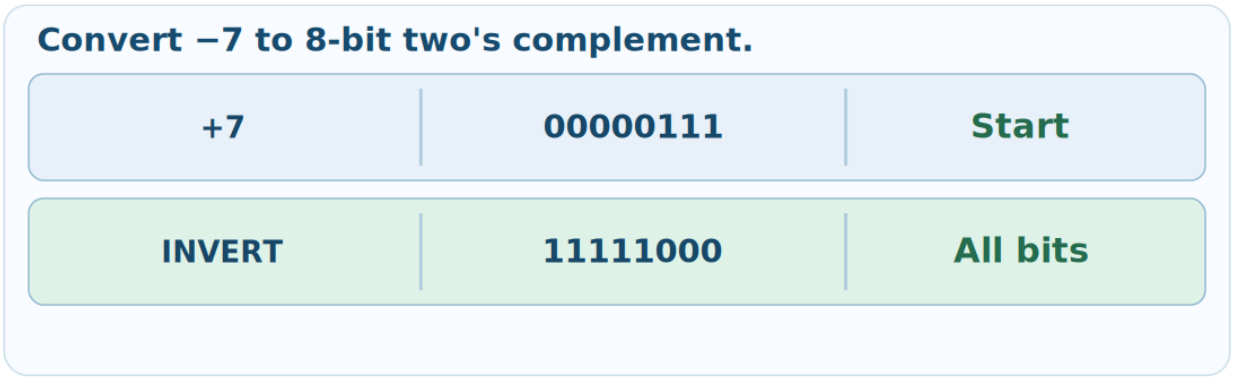
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Convert −7 to 8-bit two's complement.

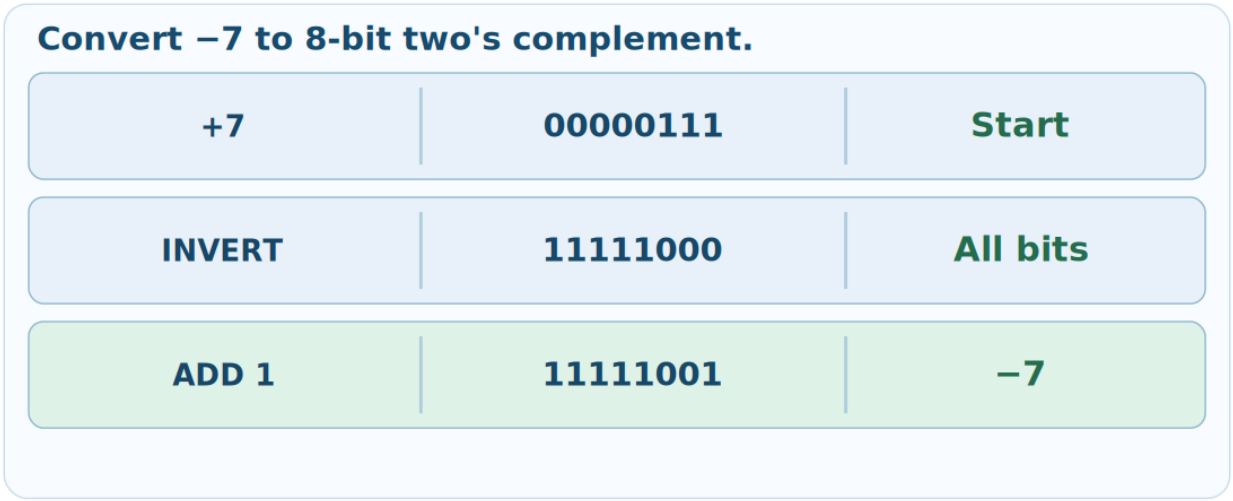
Start with +7 = 00000111 and invert all bits.



Convert −7 to 8-bit two's complement.: +7, 00000111, Start; INVERT, 11111000, All bits

Convert −7 to 8-bit two's complement.

Start with +7 = 00000111 and invert all bits.



Convert −7 to 8-bit two's complement.: +7, 00000111, Start; INVERT, 11111000, All bits; ADD 1, 11111001, −7

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain

Add one to 11111000, giving 11111001 for −7.

IMPROVE: Explain one correction to your own reasoning.